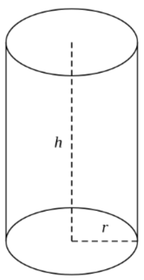
Lab 07 Inheritance and Polymorphism

Objective

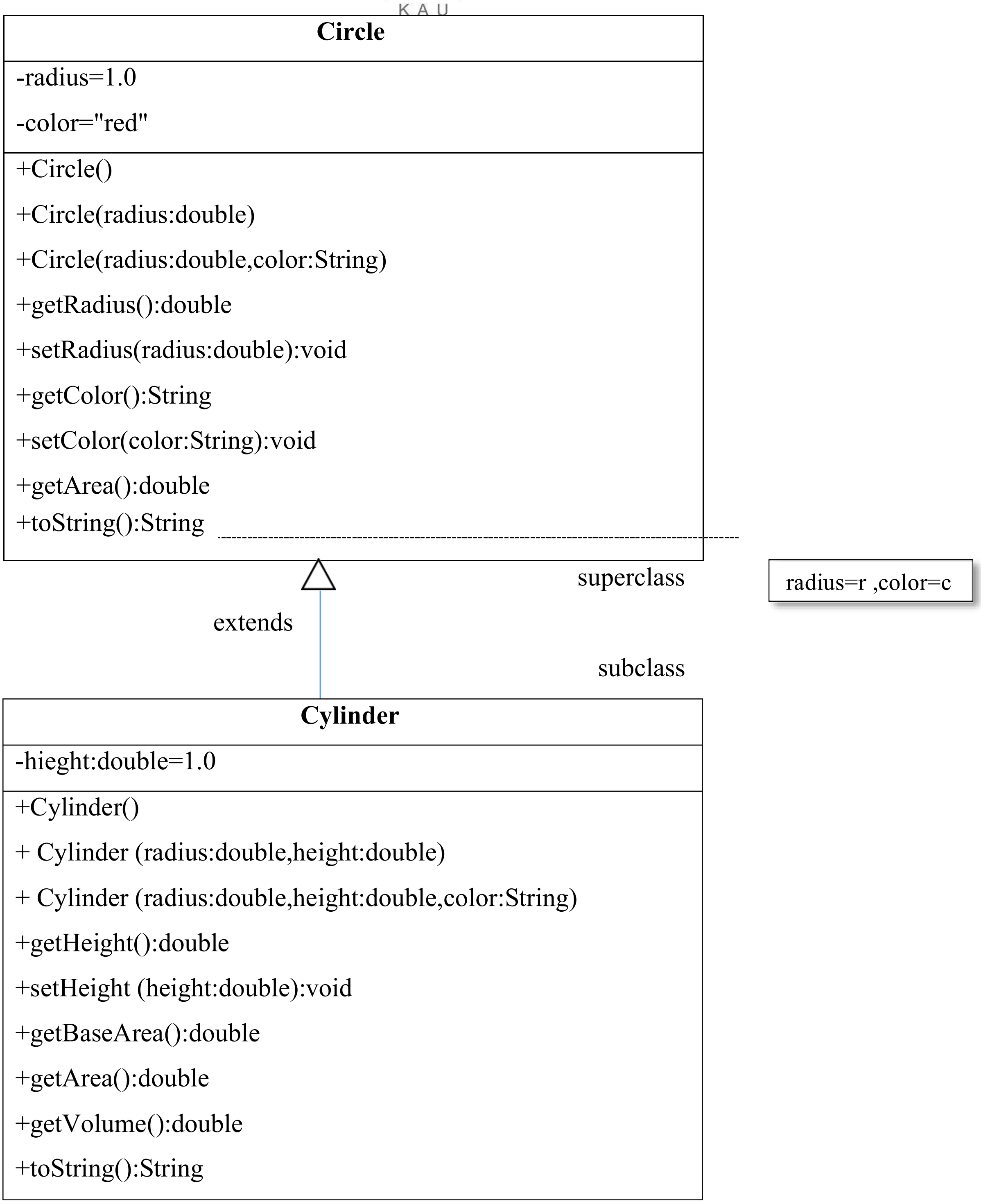
In this lab, you will learn how to:

- Using is-a relationship.
- Emphasizing on software reusability promoted by inheritance.
- Clarifying the notions of superclasses and subclasses.
- To use keyword extends to create a class that inherits attributes and behaviors from another class.
- To access superclass members with super keyword.
- Clarifying the role of constructors in inheritance hierarchies.
- Introducing the concept of method overriding.

Activity Description



As you can see, the cylinder has circles on both sides and to calculate its surface area and volume, we need first to get the area of its base. If we have a circle class already implemented, how we can implement the cylinder class by applying the inheritance concept. Make sure to follow the following UML:



Sample Output:

```
***Cylinder***
--------------
Please, enter the cylinder's base radius:5.0
--------------
Please, enter the cylinder's height:2.0
--------------
Cylinder information:
Base   radius=5.0 ,color=red ,height=2.0 , base area=78.53981633974483
Surface area=219.9114857512855 , volume=157.07963267948966
```

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Important Notes:

- Circle area= base_area=radius * radius *
- Cylinder surface area=2 _*radius*height + 2*base_area
- Cylinder volume= base_area*height
- Create Cylinder Class as described in the UML.
- Add all the required codes to complete the relationship.
- Use the super identifier to apply the inheritance relationship.
- Add the toString method to the cylinder class to display the cylinder information as shown in the sample output.
- Override the getArea method inside the Cylinder class to get the cylinder surface area instead of the circle area.
- Create Main Class and name it "TestCylinder".

code

```
// Save as "Circle.java"
public class Circle {
// private instance variables
   private double radius;
   private String color;
// Constructors
public Circle() {   this.radius = 1.0;        this.color = "red";   }
public Circle(double radius) {    this.radius = radius;     this.color = "red"; }
public Circle(double radius, String color) {this.radius = radius;   this.color = color;}

// Getters and Setters
public double getRadius() { return this.radius;     }
public String getColor() { return this.color; }
public void setRadius(double radius) { this.radius = radius; }
public void setColor(String color) {   this.color = color;    }
 //toString Method
public String toString() { return "radius=" + radius + " ,color=" + color ;    }
// Return the area of this Circle
public double getArea() {       return radius * radius * Math.PI;            } }
//---------------------------------------------------------------------------
// Save as "Cylinder.java"
public class Cylinder................. { ..................     }
//---------------------------------------------------------------------------
// save as "TestCylinder.java"
public  class  TestCylinder     { ..................    }
```